

If $f: I \subset \mathbb{R} \rightarrow \mathbb{R}$ satisfies Intermediate Value Property and for any rational $r \in \mathbb{Q}$, $f^{-1}[\{r\}]$ is closed, then f is continuous. (IVP property: If $f(a) < c < f(b)$, then $\exists x \in (a, b)$ s.t. $c = f(x)$)

Proof. Suppose that f is not continuous w.r.p. Then, there exists a sequence $\{x_n\}$ which converges to x_0 but $f(x_n)$ not converges to $f(x_0)$. That is, there exists $\varepsilon > 0$ s.t.

there exist infinitely many $n_i \in \mathbb{N}$ satisfying $|f(x_{n_i}) - f(x_0)| \geq \varepsilon$. $\left(\begin{array}{l} f(x_0) \leq f(x_{n_i}) - \varepsilon < f(x_{n_i}) \\ \text{or} \\ f(x_0) \geq f(x_{n_i}) + \varepsilon > f(x_{n_i}) \end{array} \right)$

By the density of rationals, there exist two rationals $r^- \in (f(x_0) - \varepsilon, f(x_0))$ and $r^+ \in (f(x_0), f(x_0) + \varepsilon)$.

if $f(x_0) < f(x_{n_i})$, then $f(x_0) < r^+ < f(x_{n_i})$ and

if $f(x_{n_i}) < f(x_0)$, then $f(x_{n_i}) < r^- < f(x_0)$.

By the IVP, for each n_i , there exists $t_i \in (x_0, x_{n_i}) \cup (x_{n_i}, x_0)$ s.t. $f(t_i) = r^+$ or r^- .

Since $x_{n_i} \rightarrow x_0$ as $i \rightarrow \infty$, $t_i \rightarrow x_0$ as $i \rightarrow \infty$.

For each $i \in \mathbb{N}$, $t_i \in f^{-1}[\{r^+\}] \cup f^{-1}[\{r^-\}]$, closed set, therefore the limit point x_0 satisfies

$$x_0 \in f^{-1}[\{r^+\}] \cup f^{-1}[\{r^-\}].$$

That is, $f(x_0) = r^+$ or r^- , it is contradiction.

Note: Continuous preserves the Compactness, so, a continuous map $f: X \rightarrow \mathbb{R}$ on compact X has minimum and maximum, since $f[X]$ is closed bounded.

Darboux's Theorem.

Let $f: [a, b] \rightarrow \mathbb{R}$ be a differentiable map.

Then, the derivative f' satisfies the Intermediate Value Theorem.

Proof. Suppose that $f'(a) < f'(b)$, Let α s.t. $f'(a) < \alpha < f'(b)$ be given.

Define $g(x) = f(x) - \alpha x$ ($x \in [a, b]$).

Clearly g is differentiable, and $g'(a) = f'(a) - \alpha < 0$, $g'(b) = f'(b) - \alpha > 0$.

And, since g is continuous, it has a minimum in $[a, b]$, for some c .

By the assumption, $c \neq a$ and $c \neq b$. And, since g has a minimum at c , $g'(c) = f'(c) - \alpha = 0$. So proved $\square$

Corollary. If $f: I \rightarrow \mathbb{R}$ has a primitive, then $f[I]$ is an interval.

Note: If f is defined on an interval and cannot satisfy IVT, then it has no primitive.